Ohm’s Law In Series And Parallel Circuits

Applying Kirchhoff’s junction and loop rule to calculate voltage, current, and resistance

SYNOPSIS

Ohm’s law states that $V = IR$ where V is the voltage (V), I is the current (A), and R is the resistance (Ω). Here, it says that when the pressure on the electrons is high (high voltage), the electron flow is high (large current) for a fixed bottleneck (resistance). We can write the same equation as $I = V/R$ to emphasize how voltage and resistance impact the electron flow; again, increasing voltage increases the current, i.e., current is directly proportional to voltage (or vice versa). But you can also see that increasing the resistance decreases current, i.e., the current is inversely proportional to resistance.

Series and parallel, in the water picture

The picture of water flowing through a pipe pushed by a pump is an excellent analogy whenever you think about the electric current — in place of water, you have electrons; of pipes, you have the wires; and of the pump, you have the batteries. We take this picture further to discuss, compare and contrast the current flow when circuit elements are in series or parallel.

Remember, elements **in series** come one after another sequentially, so using the water in a pipe analogy, every time an element is passed, the pressure drops, eventually reaching zero after the last element (voltages drop a bit by bit over each element until reaching zero after the last element). However, all elements see the same flow (current) since they are all part of the same circuit.

In contrast, elements **in parallel** are in “their own circuitry path,” so referring to the water in pipes analogy, elements get “their own supply” split off from the main current, wherein all elements feel the same pressure (the same voltage, because all have the same beginning and end point) while the main current is divided between different paths.

Kirchhoff’s two rules

Kirchhoff’s two rules, one for current (Junction Rule) and one for voltage (Loop Rule), allow us to calculate the currents through and voltages across the elements in series and parallel, which can be understood using the picture above. **Kirchhoff’s rules stem from the conservation laws**: the Junction Rule for charge conservation and the Loop Rule for energy conservation.

First, the Junction Rule

$$I_{\text{total, junction}} = I_1 + I_2 + I_3 + \ldots = \Sigma\, I_{\text{i}}$$

where $I_{\text{total, junction}}$ is the current before the current split-off junction and the I_{i} are the currents through elements in each path after the junction.

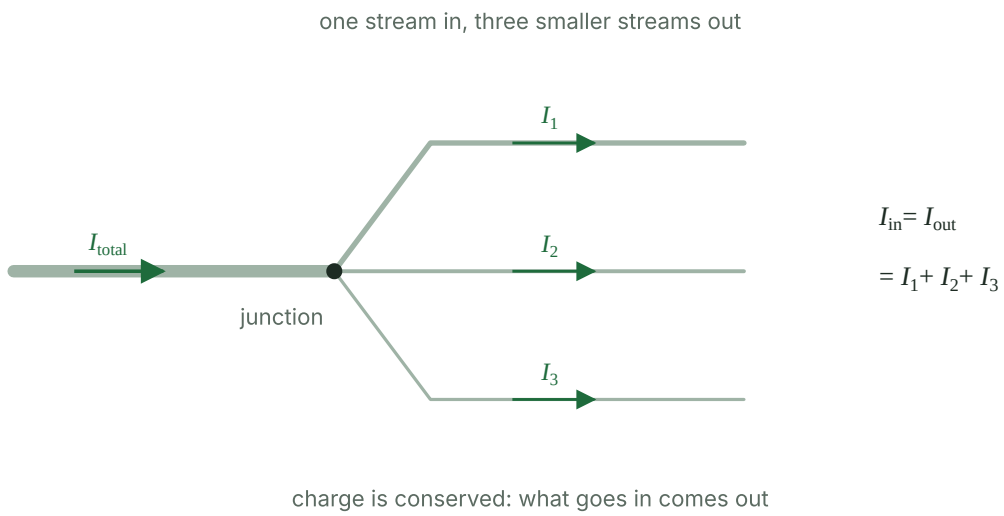


Figure 1. The current arriving at a junction is the current leaving it, added up over every branch. **The widths are drawn to scale** — the three branch streams are narrower than the trunk, and they add up to it.

In other words, $I_{in} = I_{out}$, where I_{in} is the current before the junction and I_{out} is the sum of all outgoing currents after the junction. Note that this rule applies when there is a junction, i.e., parallel circuits. This rule is nothing more than a statement that charges are conserved — we have current, which is charge flow, but the rule is fundamentally about how charges cannot be created or destroyed.

Next, the Loop Rule

$$V_{\text{battery, loop}} = V_1 + V_2 + V_3 + \ldots = \Sigma\, V_i$$

where $V_{\text{battery, loop}}$ is the battery voltage and the V_i are the voltage drops over elements in the loop. Note that this rule applies to any closed loop in which the battery and all elements are connected in series (thus the closed loop).

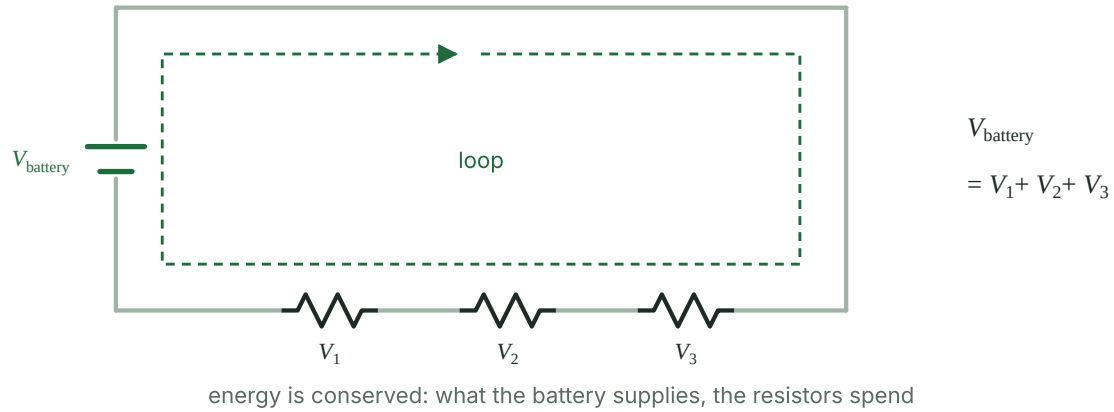


Figure 2. Round any closed loop, the battery's voltage is spent entirely across the elements in it.

In other words, the total potential (voltage) supplied by the battery is the sum of the voltage drops across all the elements in the closed circuit loop. This rule is nothing more than a statement that energy is conserved — we have potential, which is energy per charge, but the rule is fundamentally about how total supplied energy must equal the energy dissipated over all elements.

Kirchhoff's two rules lay the background for the two main equations concerning how resistances add in series and parallel circuits. And if anything, these two equations you are about to be introduced to are the take-home messages of this assignment.

Resistors in series

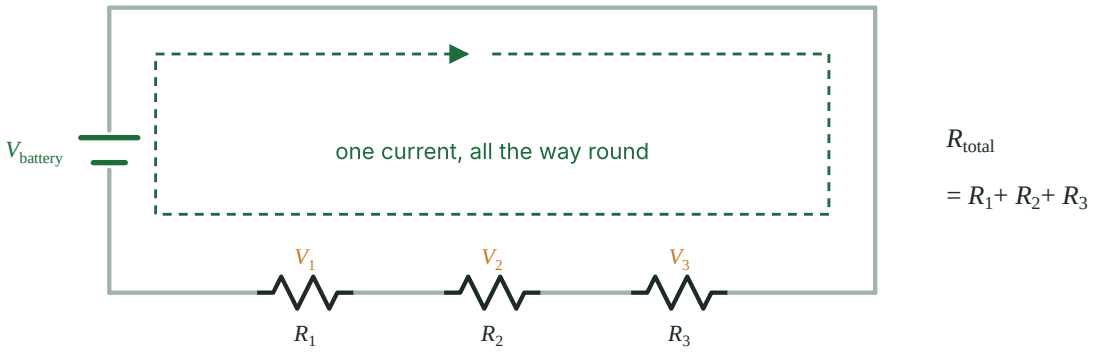


Figure 3. One loop, one current. The battery's voltage divides itself across the three resistors.

Junction rule for current: $I_{total} = I_1 = I_2 = I_3 = I$ (no split-off).
Loop rule for voltage: $V_{battery} = V_1 + V_2 + V_3$.

Resistor	Voltage (V)	Current (A)	Resistance (Ω)
1	V_1	I_1	$R_1 = V_1/I_1 = V_1/I$
2	V_2	I_2	$R_2 = V_2/I_2 = V_2/I$
3	V_3	I_3	$R_3 = V_3/I_3 = V_3/I$
Total	$V_{battery} = V_1 + V_2 + V_3$ (loop rule)	$I_{total} = I_1 = I_2 = I_3 = I$ (junction rule)	$R_{total} = V_{battery}/I_{total}$ **$R_{total} = R_1 + R_2 + R_3$**

Resistors in parallel

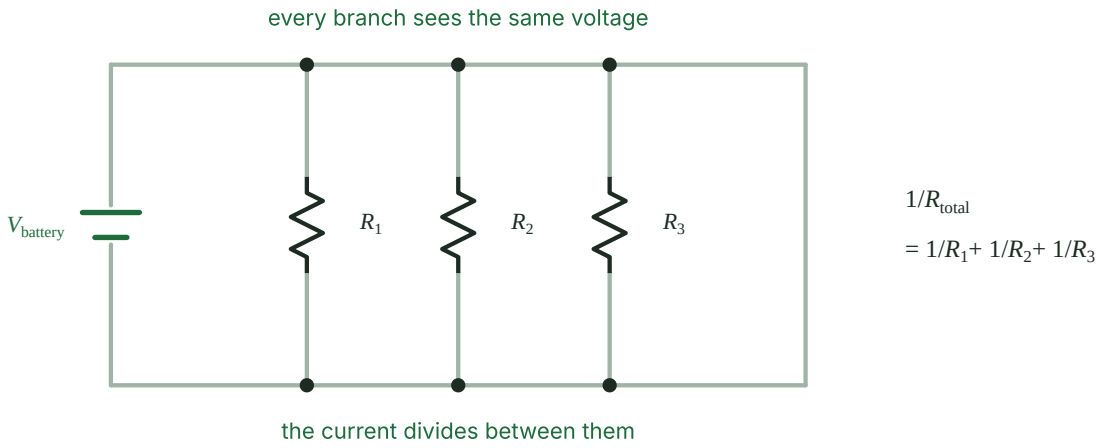


Figure 4. Three branches off the same two points. Each sees the whole battery voltage; the current is what splits.

Junction rule for current: $I_{total} = I_1 + I_2 + I_3$.
Loop rule for voltage: $V_{battery} = V_1 = V_2 = V_3 = V$.

Resistor	Voltage (V)	Current (A)	Resistance (Ω)
1	V_1	I_1	$R_1 = V/I_1$
2	V_2	I_2	$R_2 = V/I_2$
3	V_3	I_3	$R_3 = V/I_3$

Total	$V_{battery} = V_1 = V_2 = V_3 = V$ (loop rule)	$I_{total} = I_1 + I_2 + I_3$ (junction rule)	$R_{total} = V/I_{total}$ **$1/R_{total} = 1/R_1 + 1/R_2 + 1/R_3$**

THE KEY RESULTS

$$R_{total} = R_1 + R_2 + R_3 + \ldots \quad \text{in series}$$

$$1/R_{total} = 1/R_1 + 1/R_2 + 1/R_3 + \ldots \quad \text{in parallel}$$

In series, the total resistance is simply the sum of all resistors, and the same current flows through all of them, determined by $I = V_{battery} / R_{total}$.

In parallel, the inverse of the total resistance is the sum of the inverses. So when calculating the total resistance of resistors in parallel, you first add the inverses of resistance, **then invert the sum as the final step**. The same battery voltage drops across all resistors in parallel, $V = V_{battery}$, **but** the currents through them can vary with the resistance, given by $I_i = V_{battery}/R_i$.

This is a longer note than usual, but it contains essential elements to explain the last two equations above, the total resistance in series and parallel circuits. In the following, you will solve circuit problems, and there, you must first calculate the total resistance. Then, you can determine the voltages and currents for all elements.

PERFORMANCE EXPECTATIONS

1. Solve the circuit problems by determining all elements' voltages, currents, and resistance.

Instructions and Questions

Throughout: **$V_{battery} = 9$ V, $R_1 = 2$ Ω, $R_2 = 5$ Ω, $R_3 = 10$ Ω.**

1) Draw the series and parallel circuits

Draw series and parallel circuits with these three resistors.

2) The series circuit

Fill in the table for the **series** circuit. First, calculate the total resistance. Second, calculate the current in the circuit, which is the same for all resistors. Lastly, calculate the voltage drops over each element. Show work.

3) The parallel circuit

Fill in the table for the **parallel** circuit. First, calculate the total resistance. Second, calculate the total circuit current using the battery voltage and the total resistance. Lastly, calculate the currents through elements, using the voltage (V_{battery} — why?) and the resistance.

4) Confirm it in the simulation

Run the *Circuit Construction Kit: DC* PhET simulation to confirm your values in the two tables above. **Check in with Dr. Cho before moving on.**

5) Equivalent circuits — the first combination

The following circuits are **combination circuits** with both series and parallel connections. Equivalent circuits are increasingly simplified circuits in which multiple resistors in series or parallel connections are reduced to a single resistor, equaling the total resistance for either series or parallel connections. **The goal is to reduce the circuit to a single resistor.** Determine R_{12} and R_T.

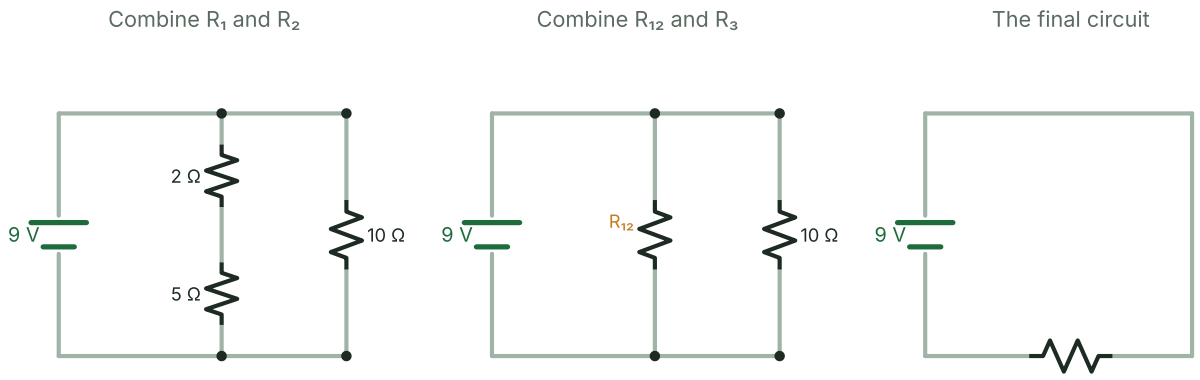


Figure 6. The circuit is drawn for you. **Work out R_{12} first, then R_T, then the current** — and say at each stage which rule you are using.

6) The table for that circuit

7) Confirm it in the simulation

Run the *Circuit Construction Kit: DC* PhET simulation to confirm your values. **Check in with Dr. Cho before moving on.**

8) Equivalent circuits — the second combination

Draw the final equivalent circuit. Determine R_{12} and R_T.

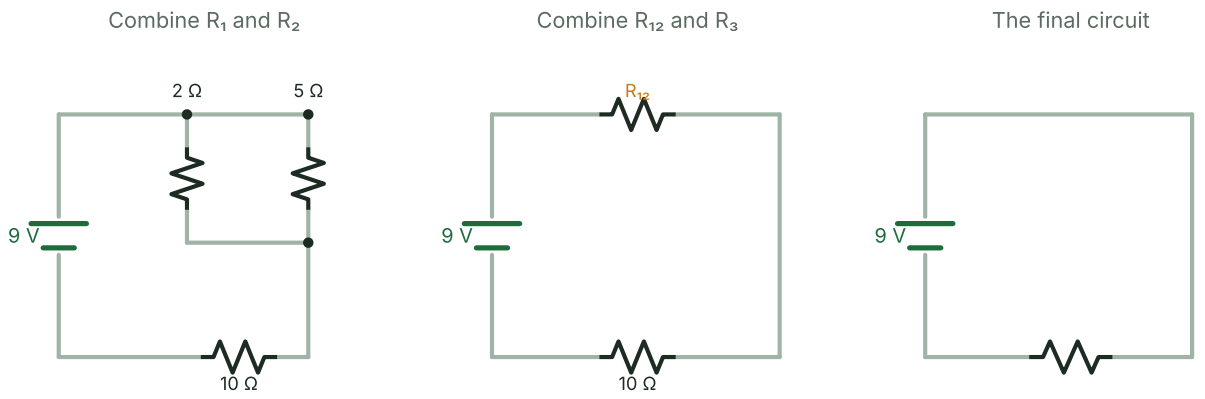


Figure 7. Same three stages, a different wiring. **Work out R_{12}, then R_T, then the current** — then say why this total is so much larger than the one in Figure 6, with the same three resistors.

9) The table for that circuit

10) Confirm it in the simulation

Run the *Circuit Construction Kit: DC* PhET simulation to confirm your values. **Check in with Dr. Cho before moving on.**

Looking ahead

Two rules, and both of them are a conservation law wearing circuit clothes: charge in equals charge out, and the energy the battery supplies is the energy the resistors spend. Everything in this note followed from those two. **The method you have just used — reduce, then work backwards — is what the rest of the module runs on.**